



# Text Mining II

## Lecture 14

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# Outline

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- Word Embedding
- Large Language Models
- Text Mining in Social Media
- Case Study



# Text Representation

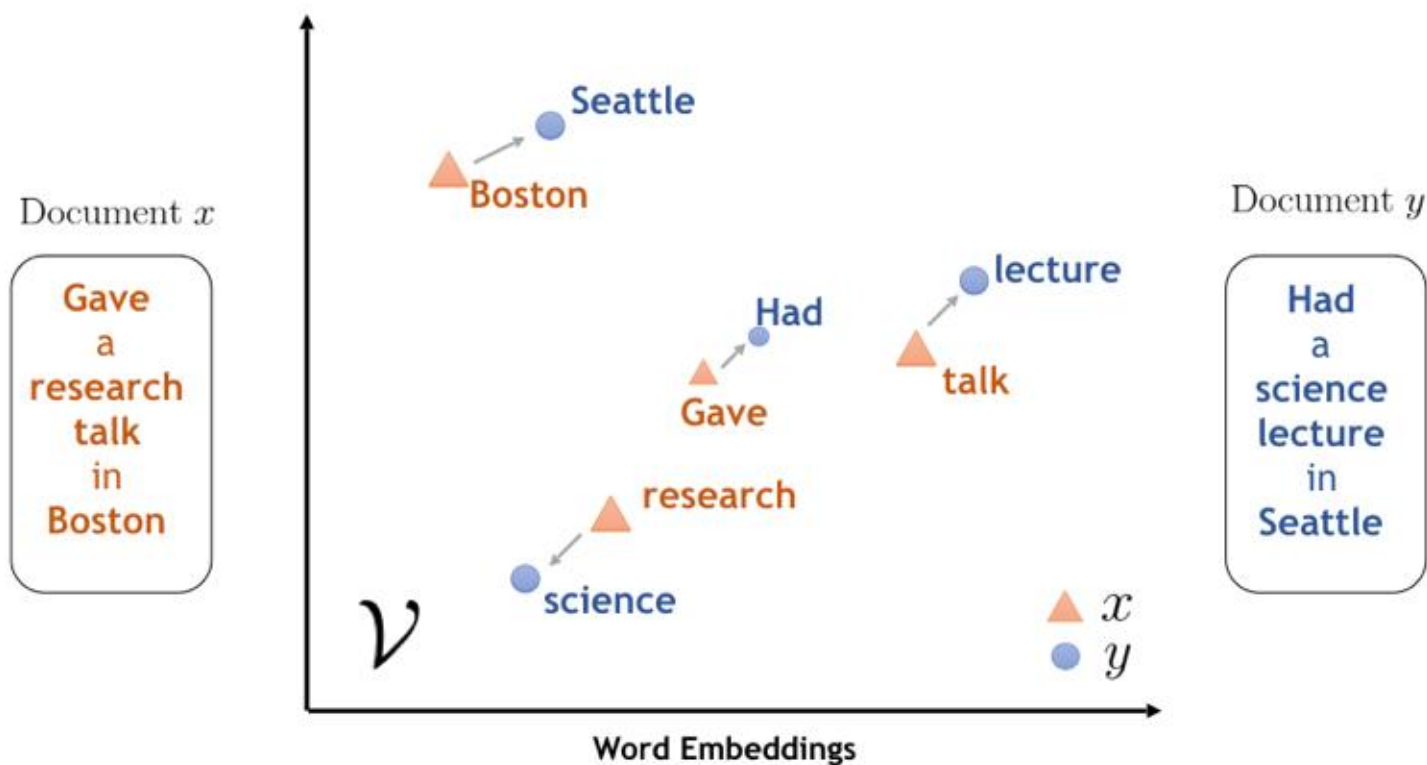
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- Text document is represented by a specific representation
  - i.e., the words (features) it contains and their occurrences
- Approach to generate attributes/document representation:
  - Vector space model (i.e., Set/Bag of words)
  - Topic model
  - **Word embedding**

# 词向量模型

## ■ 基本思想:

- 通过对词的上下文信息进行建模, 将每个词映射为一个低维连续的实数向量
- 词嵌入 (Word Embedding)



## ■ 特点:

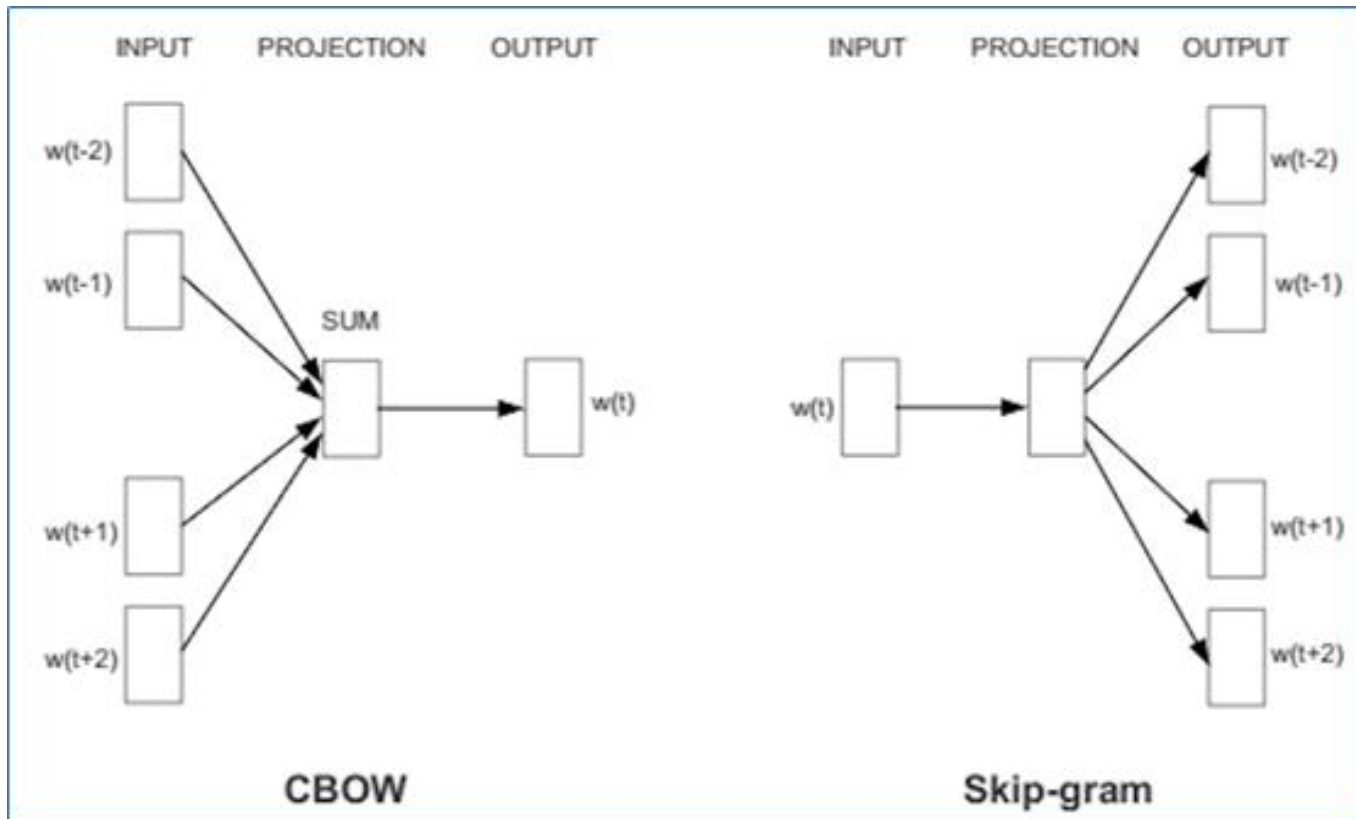
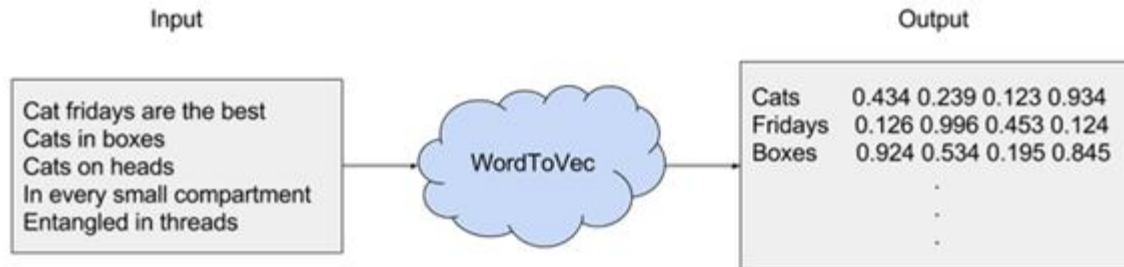
- 低维、稠密

## ■ word2vec

- 具有代表性的词向量模型之一
- Efficient Estimation of Word Representation in Vector Space by Mikolov et al. (2013)
- Continuous Bag-of-Words (CBoW)
  - ◆ 根据上下文单词预测中心词
- Continuous Skip-gram (Skip-gram)
  - ◆ 通过中心词预测上下文单词

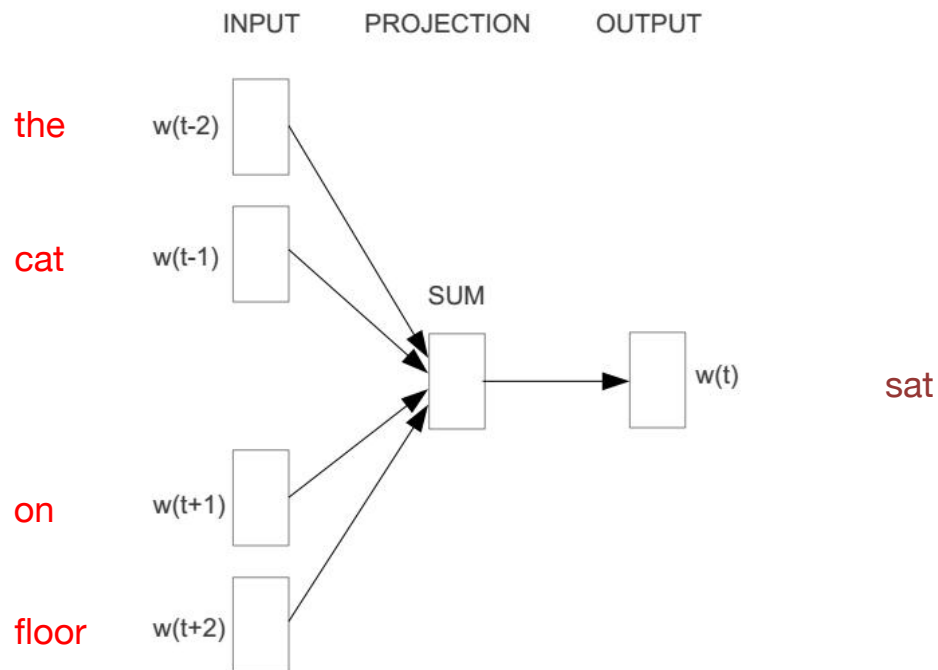
- Word2vec is not a single algorithm
- It is a software package for representing words as vectors, containing:
  - **Two distinct models:**
    - CBoW (continuous bag of words)
    - Skip-Gram
  - **Various training methods**
    - Negative Sampling
    - Hierarchical Softmax
  - **A rich preprocessing pipeline**
    - Dynamic Context Windows
    - Subsampling
    - Deleting Rare Words

# Word embedding



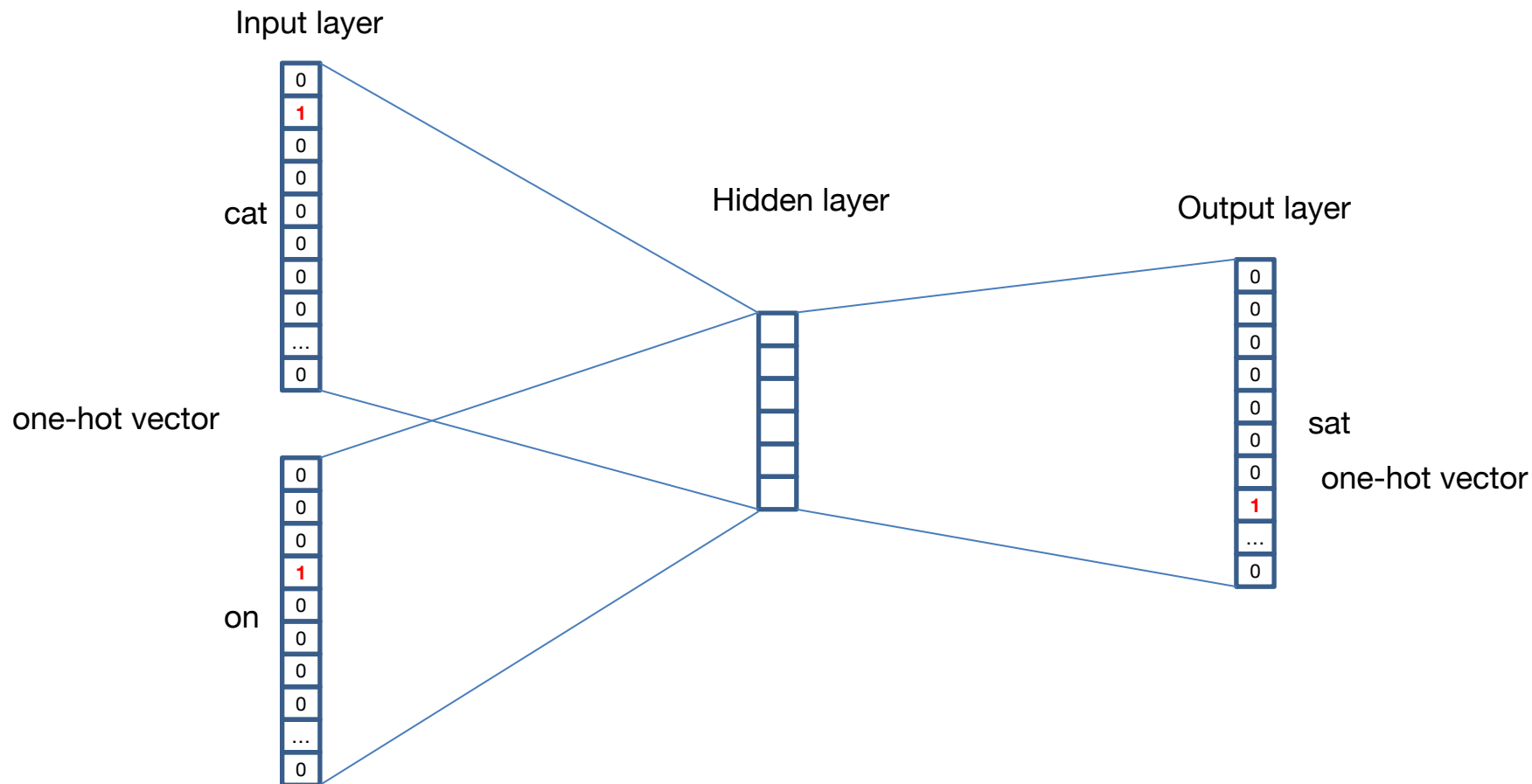
# Word2vec – CBOW

- E.g. “The cat sat on floor”
  - Window size = 2

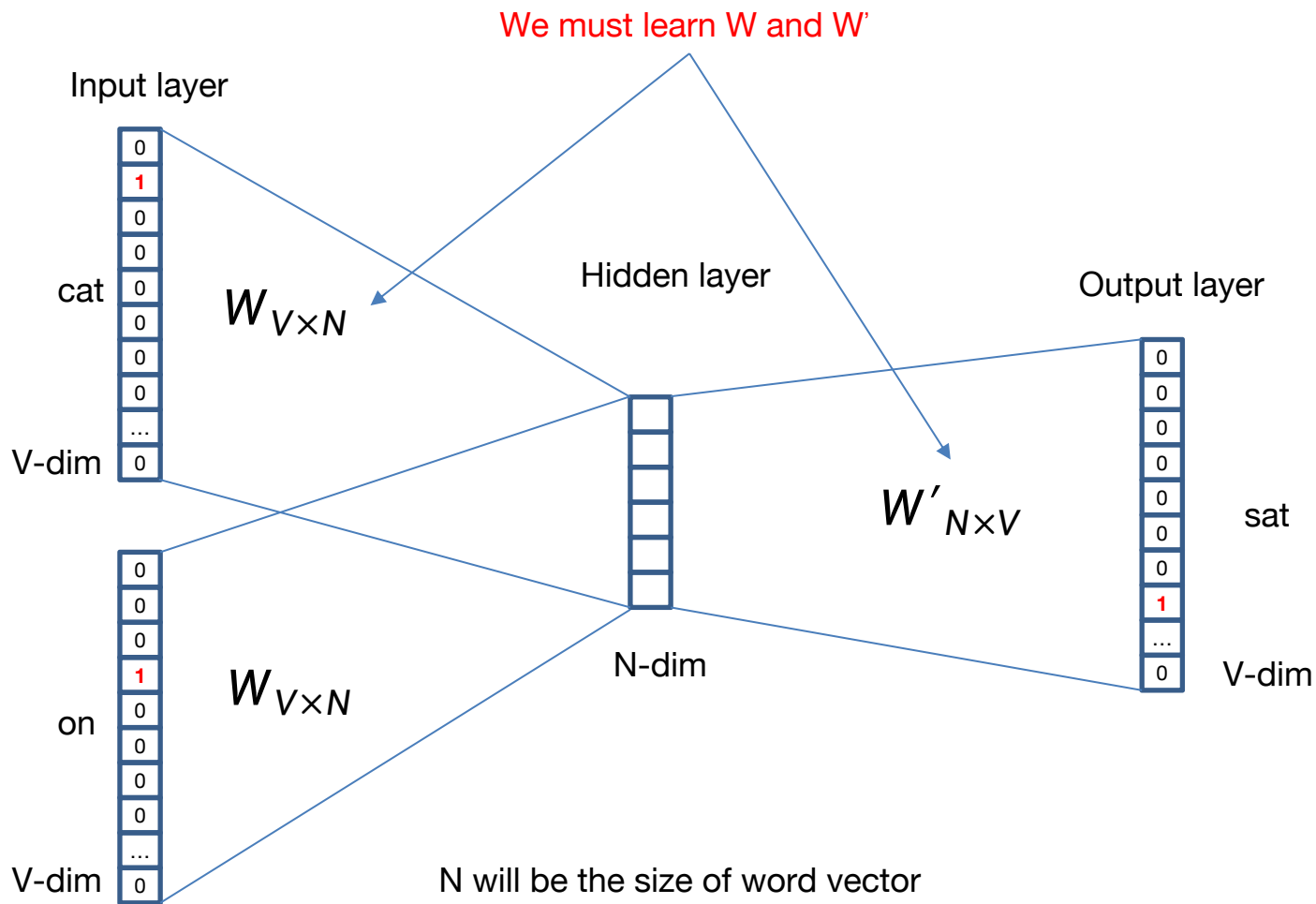


# Word2vec – CBOW

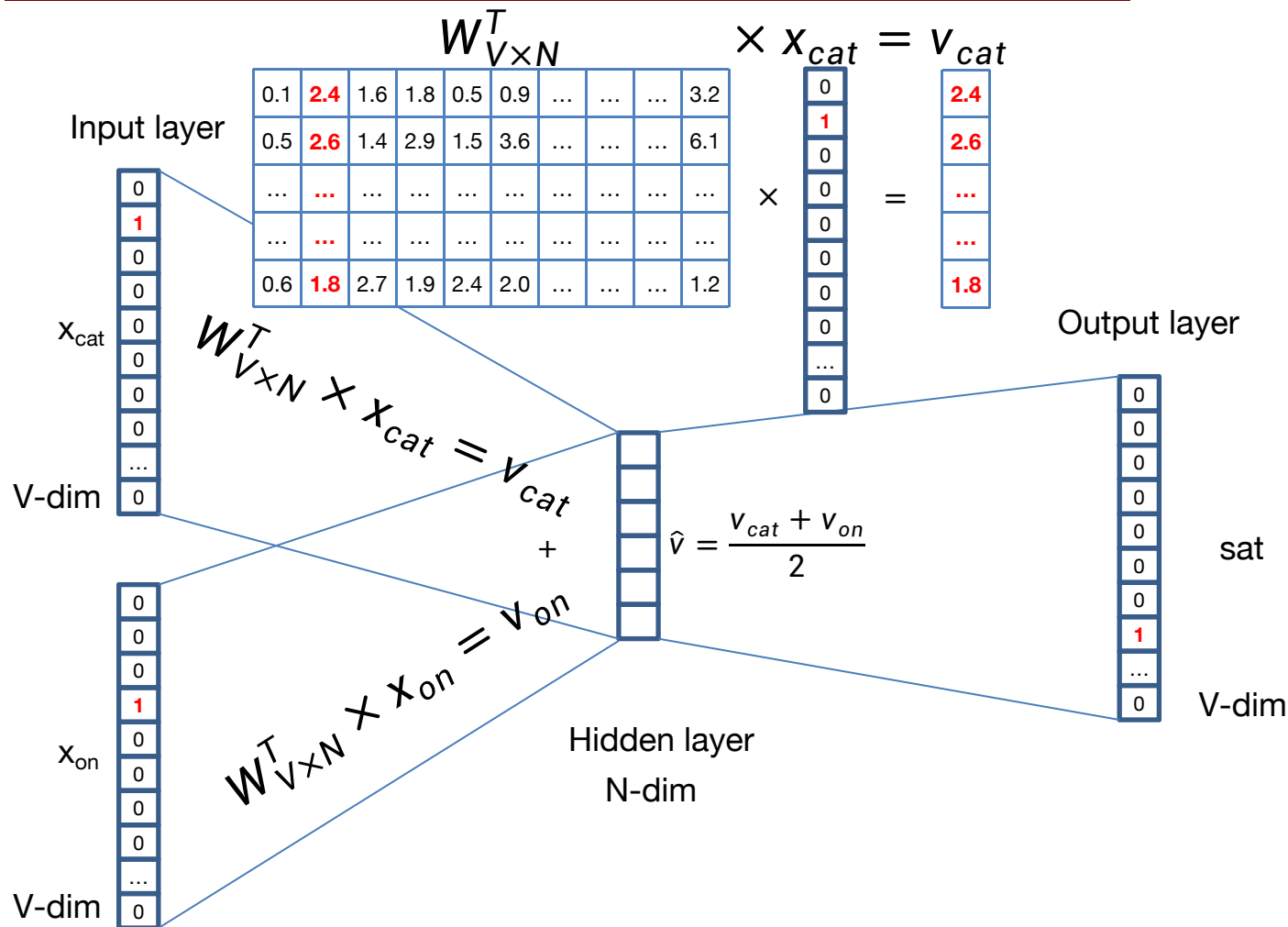
Index of cat in vocabulary



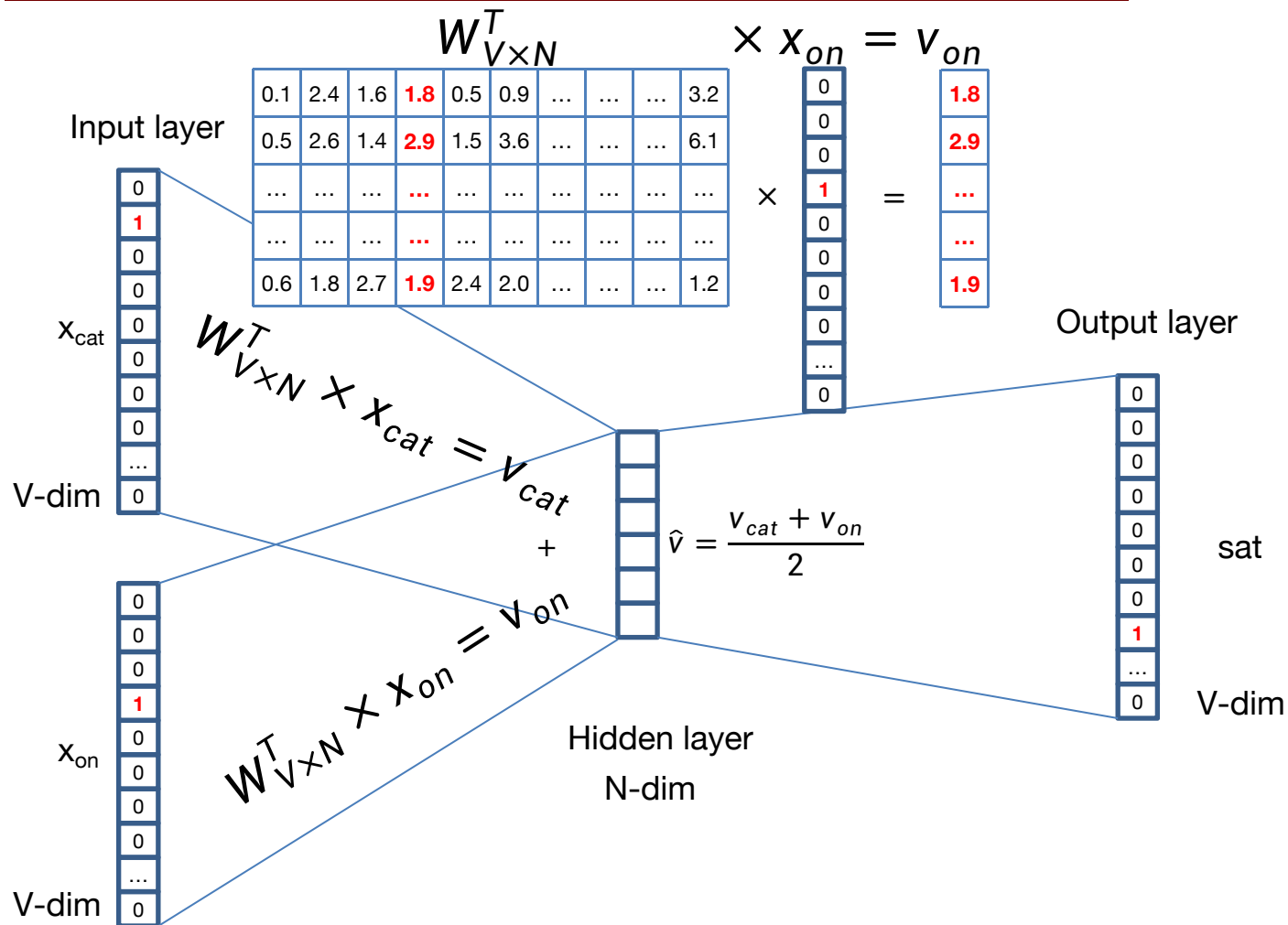
# Word2vec – CBOW



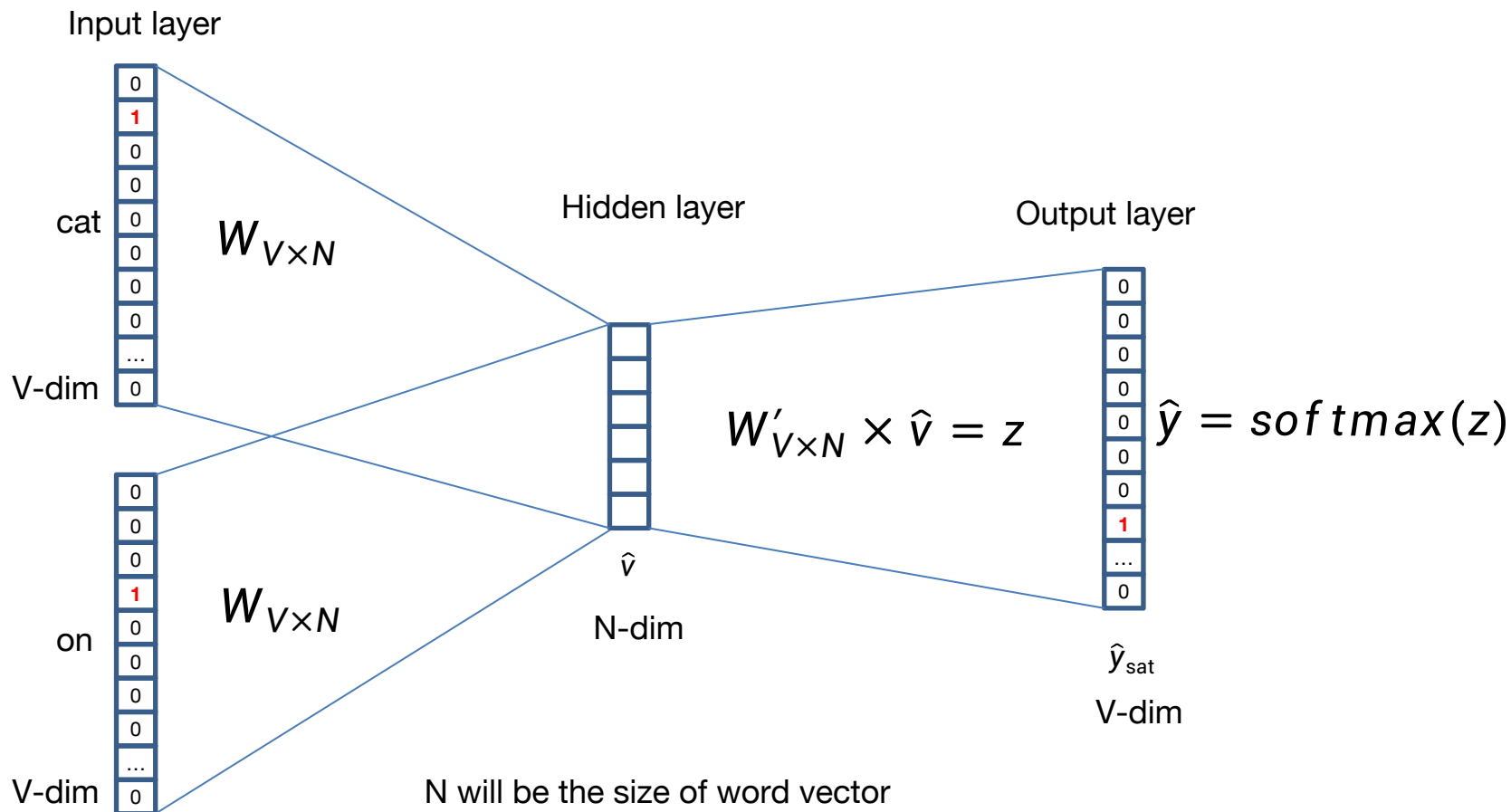
# Word2vec – CBOW



# Word2vec – CBOW

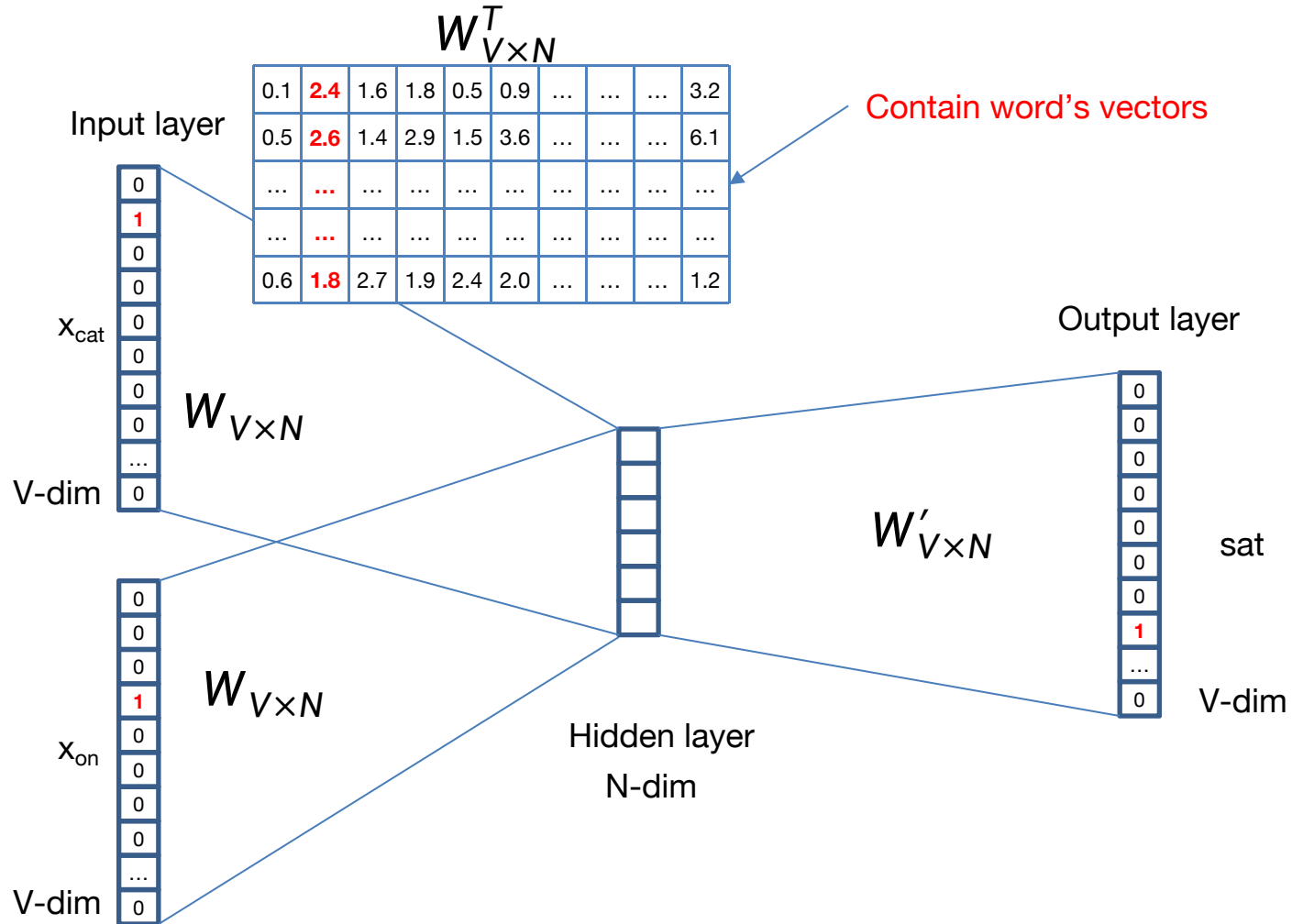


# Word2vec – CBOW





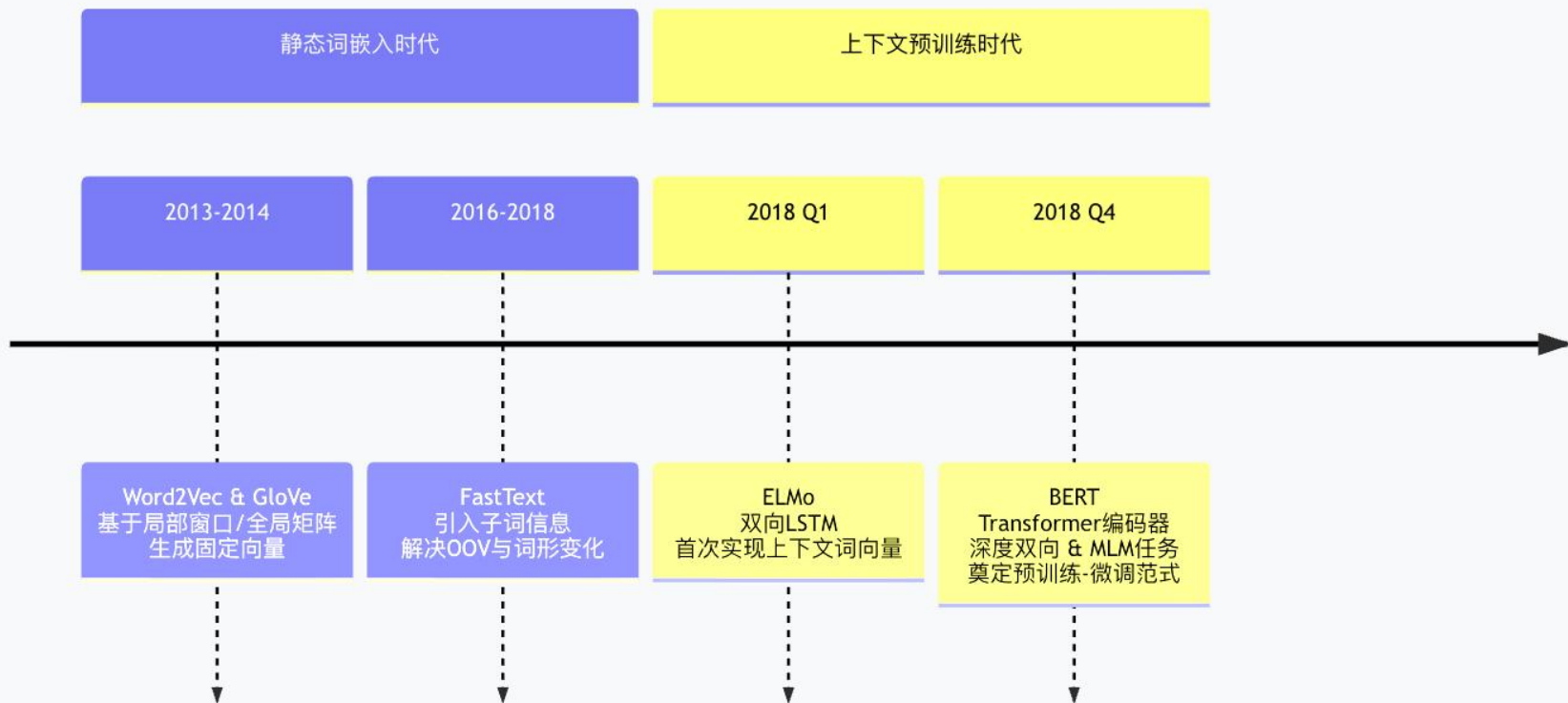
# Word2vec – CBOW



We can consider either  $W$  or  $W'$  as the word's representation.  
Or even take the average.

- **GloVe: Global Vectors for Word Representation**
  - Pennington et al. (2014)
  - <https://nlp.stanford.edu/projects/glove/>
- **FastText**
  - Mikolov et al. (2018) <Advances in Pre-Training Distributed Word Representations>
  - <https://fasttext.cc/>
- **ELMo**
  - Peters et al. (2018) <Deep Contextualized Word Representations>
  - <https://allennlp.org/elmo>
- **BERT**
  - Devlin et al. (2019) <BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding>
  - <https://arxiv.org/abs/1810.04805>

# 发展现状





# Outline

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- Word Embedding
- Large Language Models
- Case Study



写一个介绍chatgpt的文章



ChatGPT是一种基于深度学习的聊天机器人技术，它支持自然语言处理和语义理解，可以与用户进行自然语言对话。ChatGPT可以帮助公司构建聊天自动机器人，可以解答用户的提问，并且可以根据用户的问题和语境而提供自动响应，从而给用户带来友好而专业的服务。ChatGPT拥有强大和智能的语言处理技术，可以帮助企业构建智能聊天机器人，改善客户体验，提高工作效率，提供一流的客户服务。



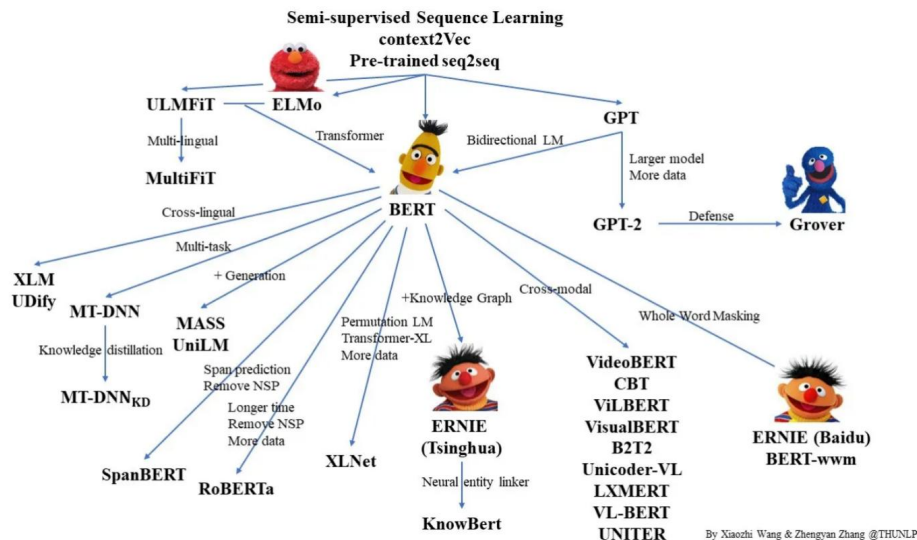
chatgpt的技术介绍



ChatGPT是一种基于深度学习的聊天机器人技术，它可以帮助企业构建智能聊天机器人，支持自然语言处理和语义理解，可以与用户进行自然语言对话，可以解答用户的提问，并且可以根据用户的问题和语境而提供自动响应。ChatGPT使用模型预测、大规模的多模态输入和多模态输出策略等技术，可以实现更为智能和准确的聊天机器人。此外，ChatGPT也具有连续领域知识的表示能力，它可以更准确地理解复杂的上下文，更好地服务于用户，提高工作效率，提供一流的客户服务。

- 语言模型本质：出现的语句是否合适合理？
- 三个发展阶段：
  - 专家语法规则模型（至 80年代）
    - 在计算机初始阶段，随着计算机编程语言的发展，归纳出的针对自然语言的语法规则。但是自然语言本身的多样性、口语化，在时间、空间上的演化，及人本身强大的纠错能力，导致语法规则急剧膨胀，不可持续。
  - 统计语言模型（至 00年）
    - 统计语言模型通过对句子的概率分布进行建模，就是计算一个句子的概率大小的模型， $P(w_1; w_2; \dots; w_m)$
  - 神经网络语言模型（till Now）
    - 词向量代替 ngram，采用连续变量(具有一定维度的实数向量)来进行单词的分布式表示，解决了维度爆炸的问题，同时通过词向量可获取词之间的相似性。
    - 通过网络的叠加和特征的逐层提取，表征除词法外，相似性，语法，语义，语用等多方面的表示。

# 预训练语言模型



预训练语言模型的脉络

经典模型：ELMO, GPT, Bert……

训练范式：

|              | Unidirectional language model | Bidirectional language model                                                                              | Sequence-to-sequence model           |
|--------------|-------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------|
| Architecture | Transformer decoder           | Transformer encoder                                                                                       | Transformer                          |
| Pre-training | Language modeling (2)         | Mask language modeling (3)                                                                                | Sequence-to-sequence learning        |
| Tasks        | Language generation           | Language understanding                                                                                    | Sequence-to-sequence                 |
| Models       | GPTs <sup>3,25,26</sup>       | BERT, <sup>8</sup> RoBERTa, <sup>17</sup> ALBERT, <sup>14</sup> XLNet, <sup>36</sup> Electra <sup>7</sup> | BART, <sup>15</sup> T5 <sup>24</sup> |

# 预训练语言模型 -> 大语言模型

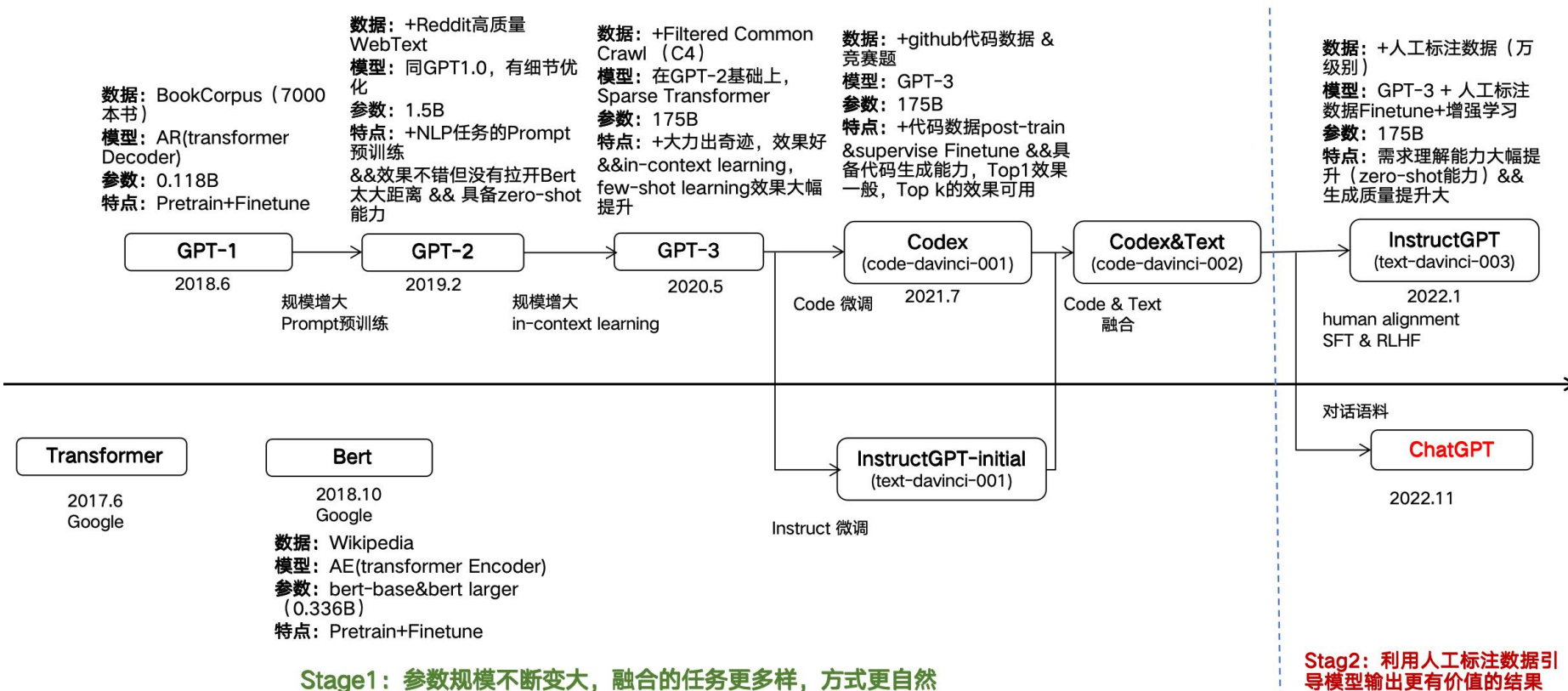
- ▶ **大规模语言模型**：通常指参数量超过 10B 的模型
  - ▶ 更多的计算量、推理开销更大
  - ▶ 泛化性能更强，出现涌现能力

|          | 预训练语言模型<br>(小模型、常规模型) | 大规模生成式语言模型          |
|----------|-----------------------|---------------------|
| 典型模型     | ELMo, BERT, GPT-2     | GPT-3、ChatGPT、LLaMA |
| 模型结构     | BiLSTM, Transformer   | Transformer         |
| 注意力机制    | 双向、单向                 | 单向                  |
| 训练方式     | 去噪自编码模型               | 自回归生成               |
| 擅长任务类型   | 理解、判断                 | 生成                  |
| 模型规模     | 1-10亿级参数              | 10-1000亿级参数         |
| 下游任务应用方式 | 微调                    | 微调 & 提示学习           |
| 涌现能力     | 小数据领域迁移               | 上下文学习，思维链提示         |

# 本质变化

- ▶ 量变到质变：从过拟合（overfitting）风险到欠拟合（underfitting）风险
- ▶ 训练数据的变化：多元化
  - ▶ 不再仅仅是自然语言文本，而是多种数据的组合：自然语言文本、编程代码、化学分子式，乃至基因序列，甚至图像
- ▶ 训练方式的变化：从判别式预训练（BERT为典型）全面转向生成式预训练（GPT为典型）
- ▶ 模型架构的变化：从双向Transformer转向单向Transformer（Decoder-only）
- ▶ 应用方式的变化：从微调走向更为友好的提示学习
  - ▶ 样本更少，从必须一定的标注样本，到少样本，乃至零样本
  - ▶ 提示学习的工作形式逼近人机对话形式

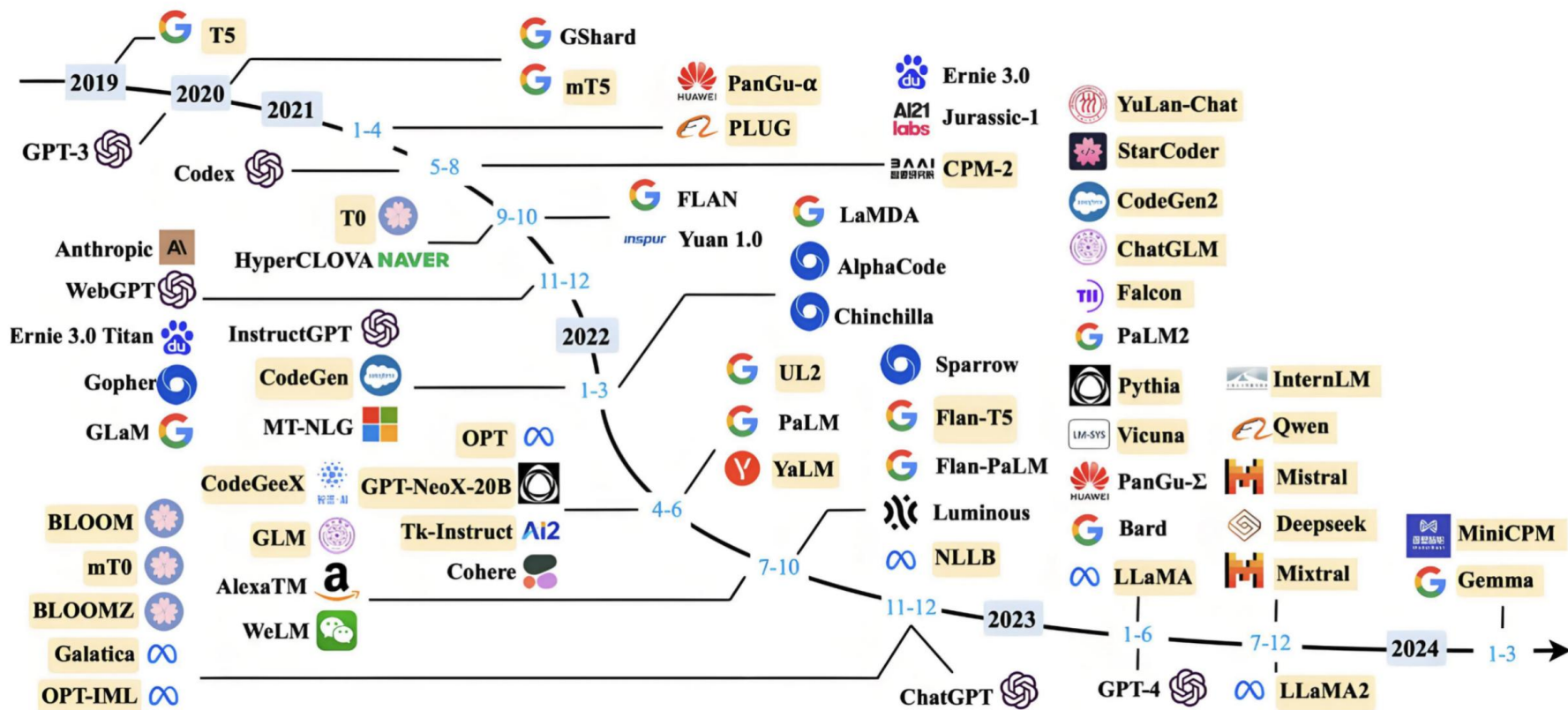
# ChatGPT 的演进概览



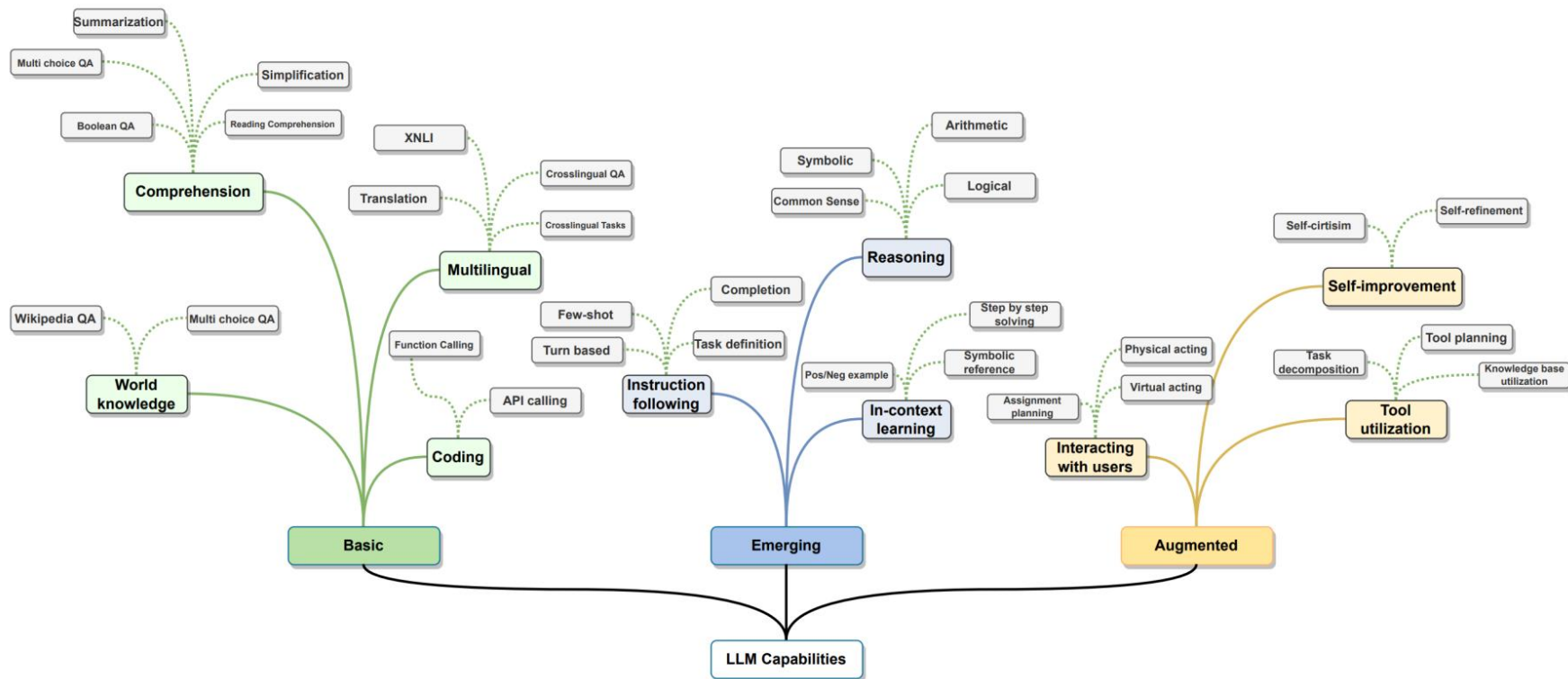
# 大语言模型的进化

| 模型          | 机构        | 参数量           | 训练数据量                                   | 方法和结论                                                                                                                                    | 文献                                                                         |
|-------------|-----------|---------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| GPT3        | OpenAI    | 0.1B~175B     | 约500B tokens                            | Transformer Decoder                                                                                                                      | Language Models are Few-Shot Learners                                      |
| LaMDA       | Google    | 137B          | 1.56T words                             | Transformer Decoder三大目标：质量、安全和根基性（事实正确性）。质量分为合理性、特异性和趣味性；主要根据以上评测指标来约束生成，将生成和排序融合到一起，同时增加了两个任务来融入知识（输入对话上下文，输出知识查询语句；输入知识查询语句，输出生成的最终结果） | LaMDA: Language Models for Dialog Applications                             |
| WebGPT      | Open AI   | 760M、13B、175B | Demonstraions: 6209<br>Comprisons:21548 | 其核心思想是使用GPT3模型强大的生成能力，学习人类使用搜索引擎的一系列行为，通过训练奖励模型来预测人类的偏好，使WebGPT可以自己搜索网页来回答开放域的问题，而产生的答案尽可能满足人类的喜好。                                       | WebGPT: Browser-assisted question-answering with human feedback            |
| Sparrow     | Deep Mind | 70B           | /                                       | 核心为从人类反馈中学习，创造更安全的对话助手。                                                                                                                  | Improving alignment of dialogue agents via targeted human judgements       |
| FLAN-T5     | Google    | 540B          | 1800个任务                                 | 任务的指令 与数据进行拼接。统一的输入输出格式（4种类型），引入chain-of-thought，大幅提高任务数量，大幅提高模型体积；                                                                      | Scaling Instruction-Finetuned Language Models                              |
| Gopher      | Deep Mind | 44M~ 280B     | 10.5TB                                  | 堆参数的大模型                                                                                                                                  | Scaling Language Models: Methods, Analysis & Insights from Training Gopher |
| PaLM        | Google    | 8B、62B、540B   | 780B tokens 包括网页、书籍、维基百科、代码、社交对话        | Transformer Decoder                                                                                                                      | PaLM: Scaling Language Modeling with Pathways                              |
| InstructGPT | Open AI   | 1.3B、6B、175B  | 微调数据1w+，Reward Model 4w+，PPO无标注数据4w+    | GPT3 Finetune+RLHF指令微调                                                                                                                   | Training language models to follow instructions with human feedback        |
| ChatGPT     | Open AI   | /             | 推测和InstructGPT差不多                       | GPT3.5（codex基础上）Finetune+RLHF+解决对齐问题                                                                                                     | /                                                                          |

# 大语言模型的产业实践



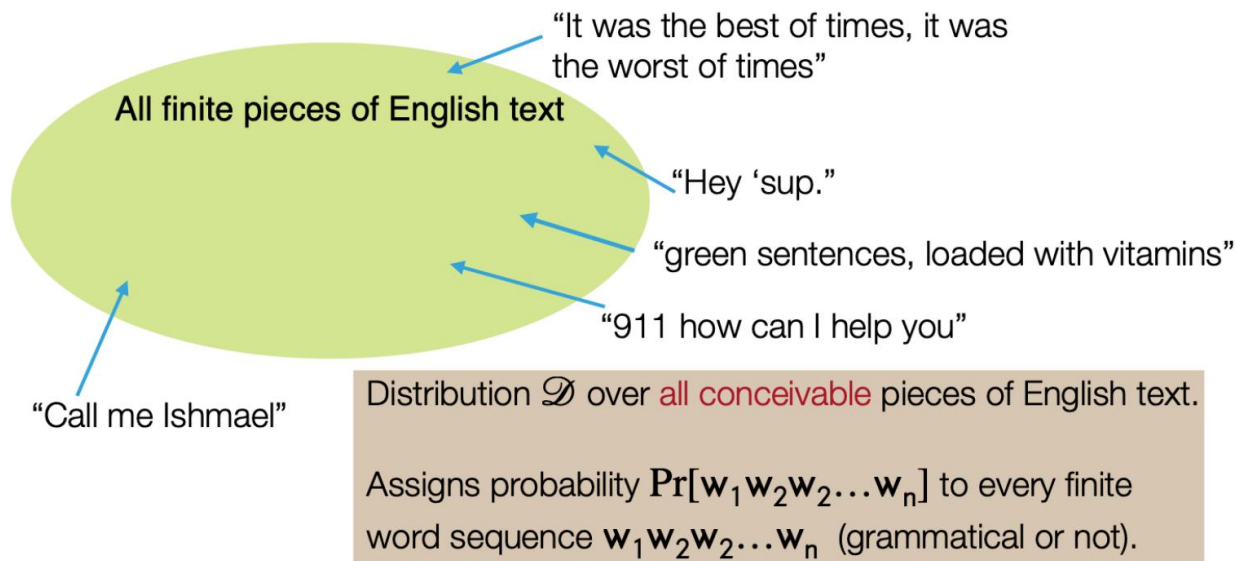
# 大语言模型的能力版图



Shervin Minaee, Tomas Mikolov, Narjes Nikzad, Meysam Chenaghlu, Richard Socher, Xavier Amatriain, Jianfeng Gao. Large Language Models: A Survey. <https://arxiv.org/pdf/2402.06196.pdf>

# Language models

- A probabilistic model that assigns a probability  $P[w_1, w_2, \dots, w_n]$  to every finite sequence  $w_1, \dots, w_n$  (grammatical or not)



# Language models

Conditional probability

$$p(w_1, w_2, w_3, \dots, w_N) = p(w_1) p(w_2|w_1) p(w_3|w_1, w_2) \times \dots \times p(w_N|w_1, w_2, \dots, w_{N-1})$$

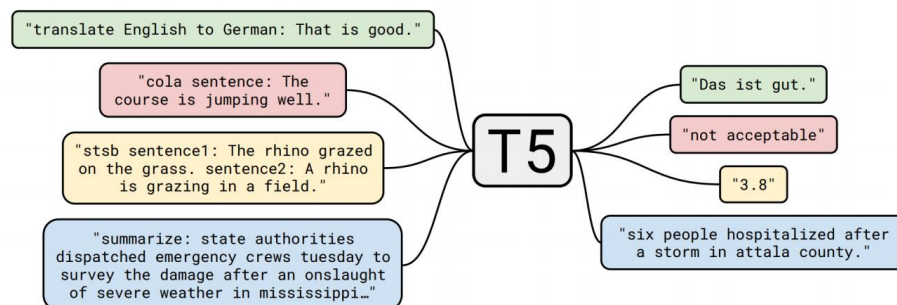
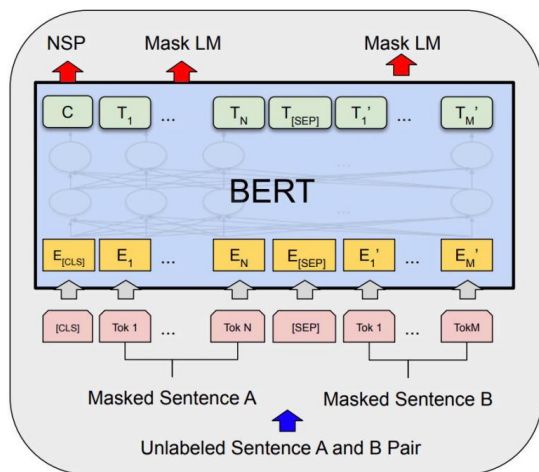
Sentence: “the cat sat on the mat”

$$\begin{aligned} P(\text{the cat sat on the mat}) = & P(\text{the}) * P(\text{cat}|\text{the}) * P(\text{sat}|\text{the cat}) \\ & * P(\text{on}|\text{the cat sat}) * P(\text{the}|\text{the cat sat on}) \\ & * P(\text{mat}|\text{the cat sat on the}) \end{aligned}$$

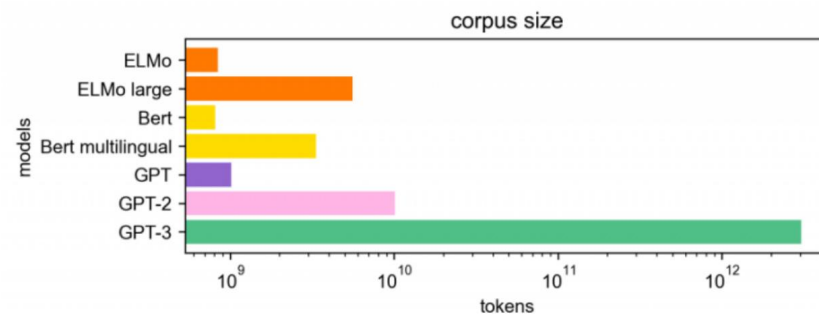
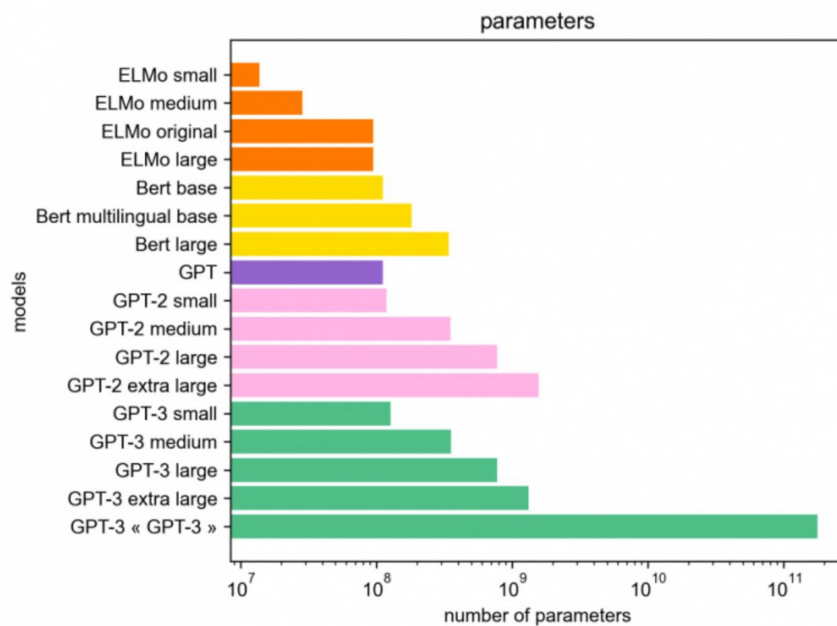
Implicit order

# Language models

- Decoder-only models (GPT-x models)
- Encoder-only models (BERT, RoBERTa, ELECTRA)
- Encoder-decoder models (T5, BART)



# Large language models

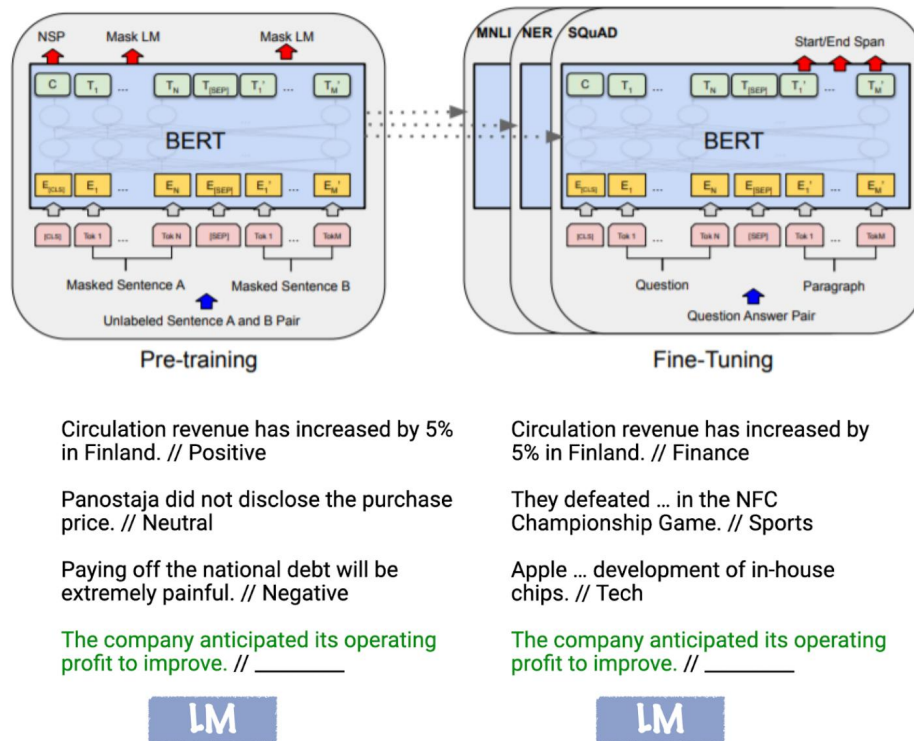


More recent models: PaLM (540B), OPT (175B), BLOOM (176B)...

Image source: <https://hellofuture.orange.com/en/the-gpt-3-language-model-revolution-or-evolution/>

# Pre-training and adaptation

- **Pre-training:** trained on huge amounts of unlabeled text using “self-supervised” training objectives
- **Adaptation:** how to use a pre-trained model for your downstream task?
  - What types of NLP tasks (input and output formats)?
  - How many annotated examples do you have?

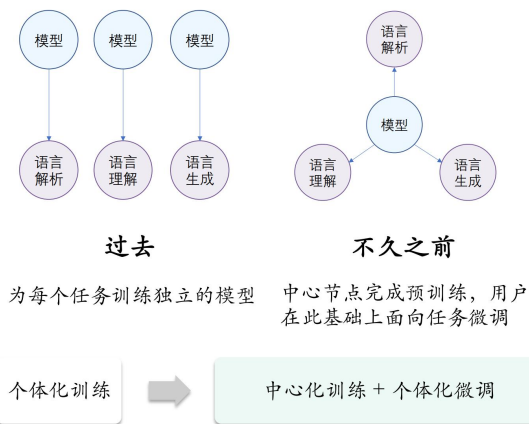
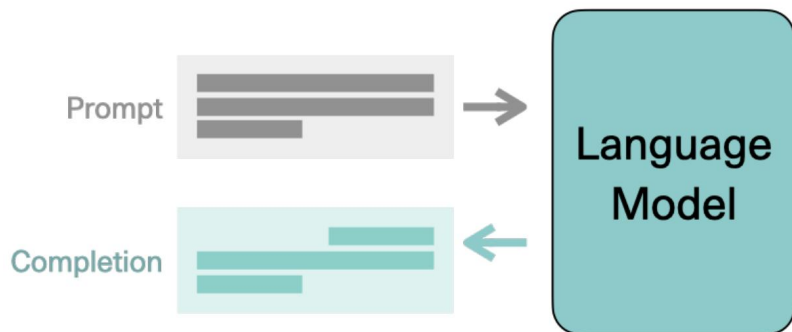


<https://ai.stanford.edu/blog/understanding-incontext/>



# Features of LLMs

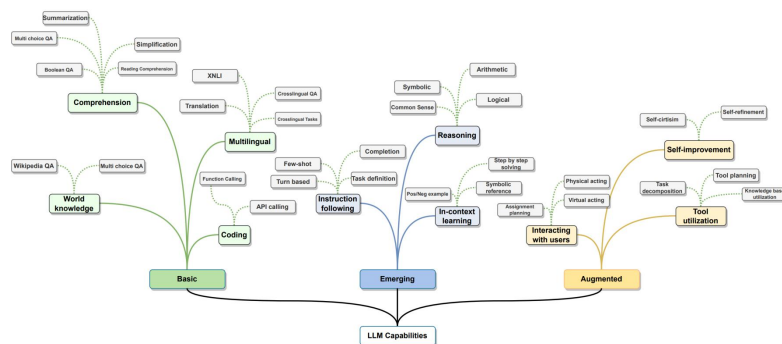
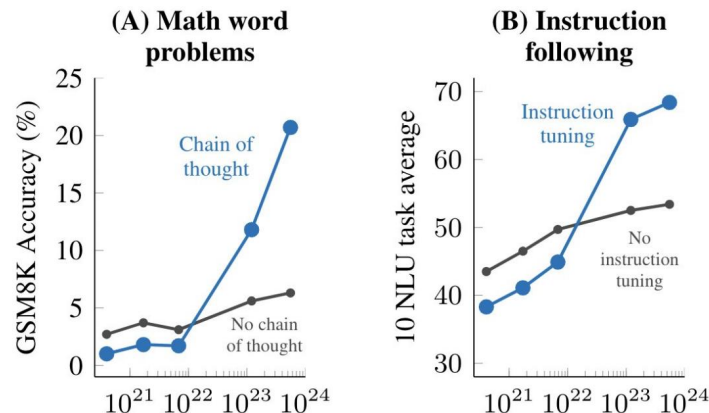
- The promise: one single model to solve many NLP tasks



现在 (大规模语言模型)

- 提示学习
  - 上下文学习
  - 思维链提示
- 轻量化微调

- Emergent properties in LLMs



Shervin Minaee, Tomas Mikolov, Narjes Nikzad, Meysam Chenaghlu, Richard Socher, Xavier Amatriain, Jianfeng Gao. Large Language Models: A Survey. <https://arxiv.org/pdf/2402.06196.pdf>



# Outline

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- What is Text Mining
- Text Data Preprocess
- Text Representation
- Text Mining Tasks
- Large Language Models
- Case Study



# Outline

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- What is Text Mining
- Text Data Preprocess
- Text Representation
- Text Mining Tasks
- Case Study

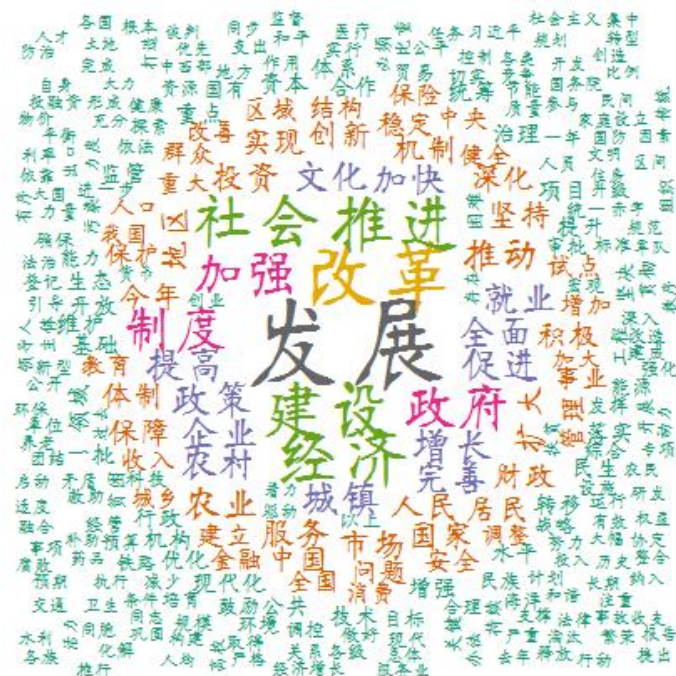


# Some Text Mining Tasks

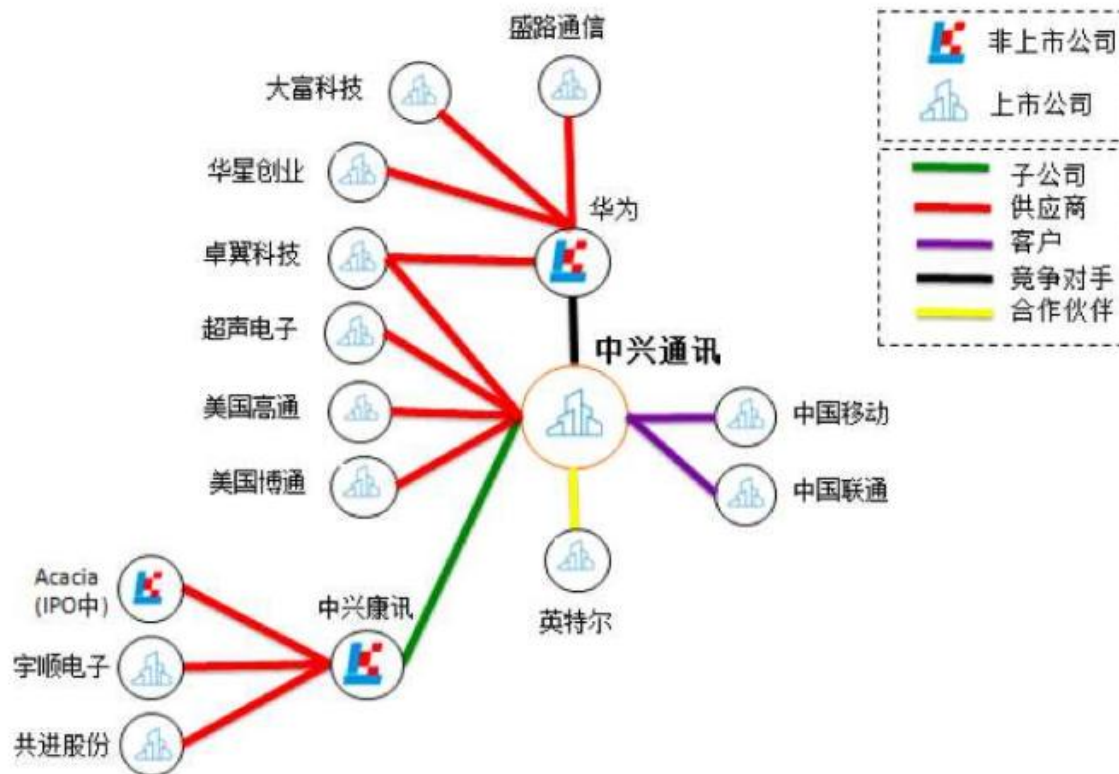
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- Visualization
- Data Mining Techniques
  - Classification
  - Clustering
  - Event Detection
  - Collaborative Question Answering

- **Data Pre-process and Visualization**
  - Word Cloud (词云)
  - Knowledge Graph (知识图谱)
- **Text Mining Results and Visualization**
  - Knowledge Graph (知识图谱)
  - General Data Visualization Tools



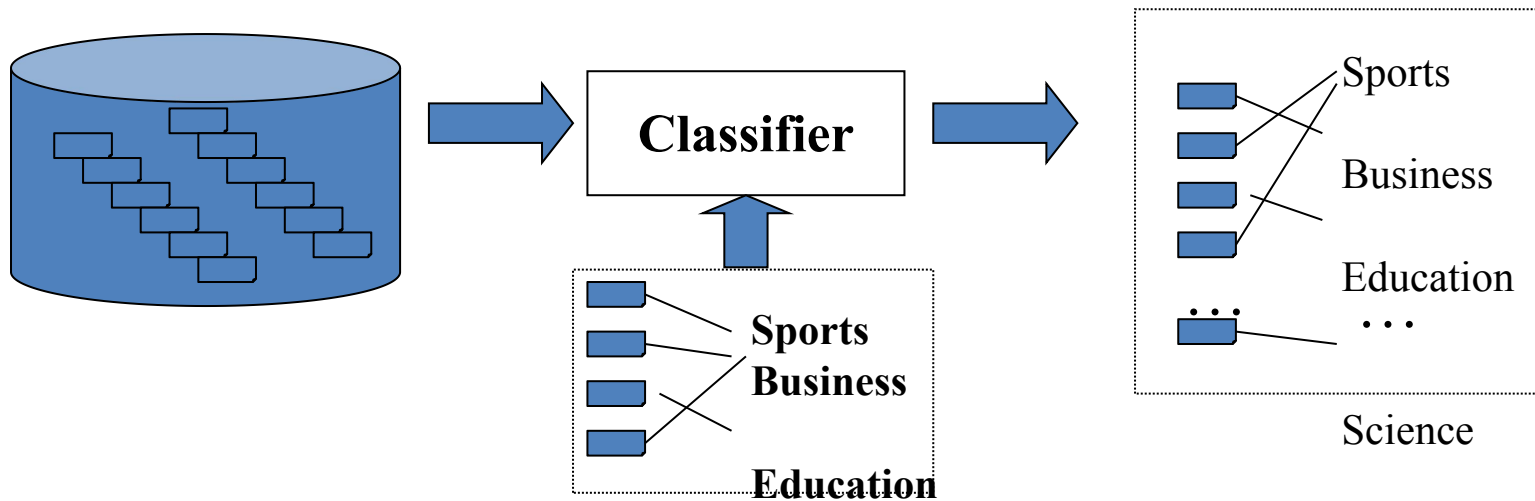
# Text Mining And Visualization



Knowledge Graph

# Document classification

- Pre-given documents and labeled document examples
- Classify new documents
- A standard classification (supervised learning) problem



# Document and clustering

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- Clustering
  - k-means
  - DBSCAN
  - ...



娱乐类



时政类



财经类

# Event Detection

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- Event Detection aims to monitor a data source and detect the occurrence of an event that is captured within that source
  - Monitor Real-Time Events via Social Media
  - Example: Detecting earthquake when people are posting live-situation through microblogging like Twitter & Facebook
- Improve traditional news detection
  - Large number of news are generated from various new channels, but only few receive attention from users
  - Researchers proposed to utilize blogosphere to facilitate news detection



# Collaborative Question Answering

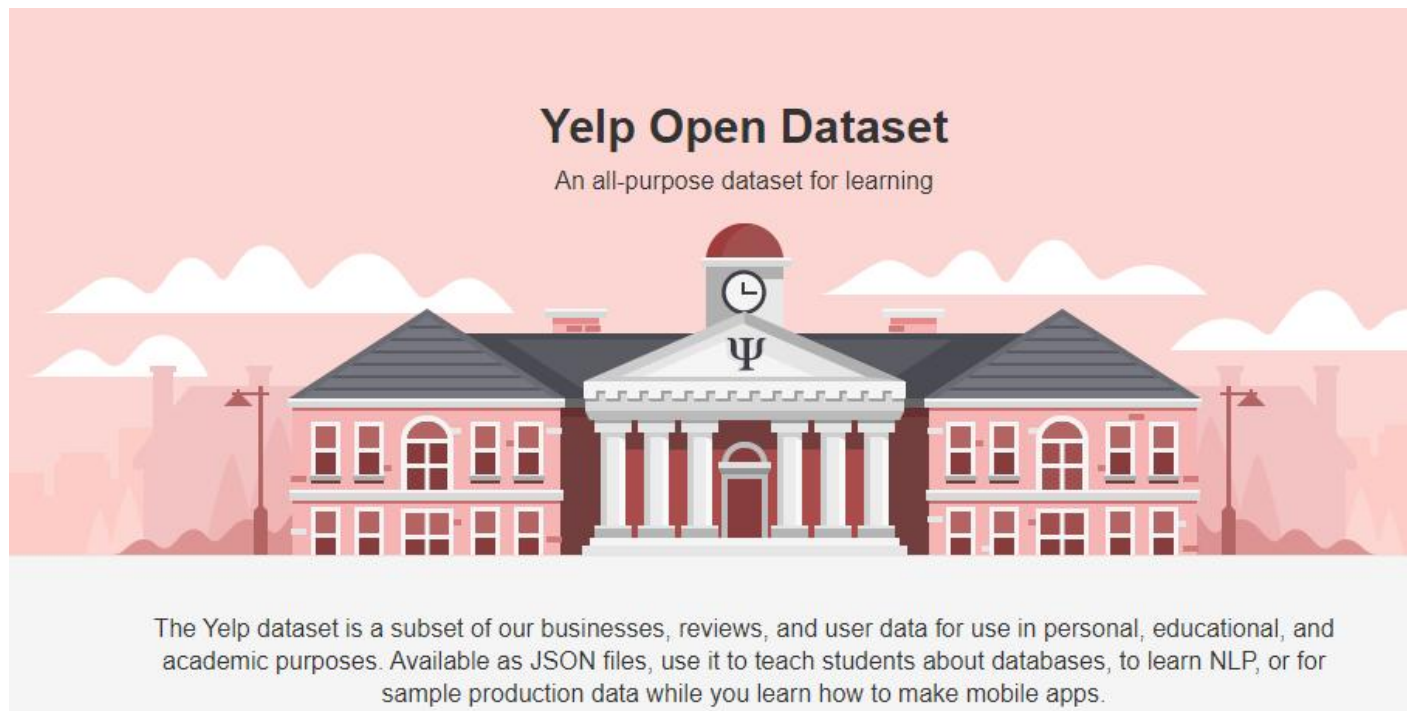
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- Collaborative question answering services bring together a network of self-declared “experts” to answer questions posted by other people
- Through text mining, a tremendous amount of historical QA pairs have built up their databases, and this transformation gives users an alternative place to look for information, as opposed to a web search
- The corresponding best solutions could be explicitly extracted and returned



# Case Study

- <https://www.yelp.com/dataset>



# Case Study

- <https://www.yelp.com/dataset>

## The Dataset



8,635,403 reviews



160,585 businesses



200,000 pictures



8 metropolitan areas

1,162,119 tips by 2,189,457 users

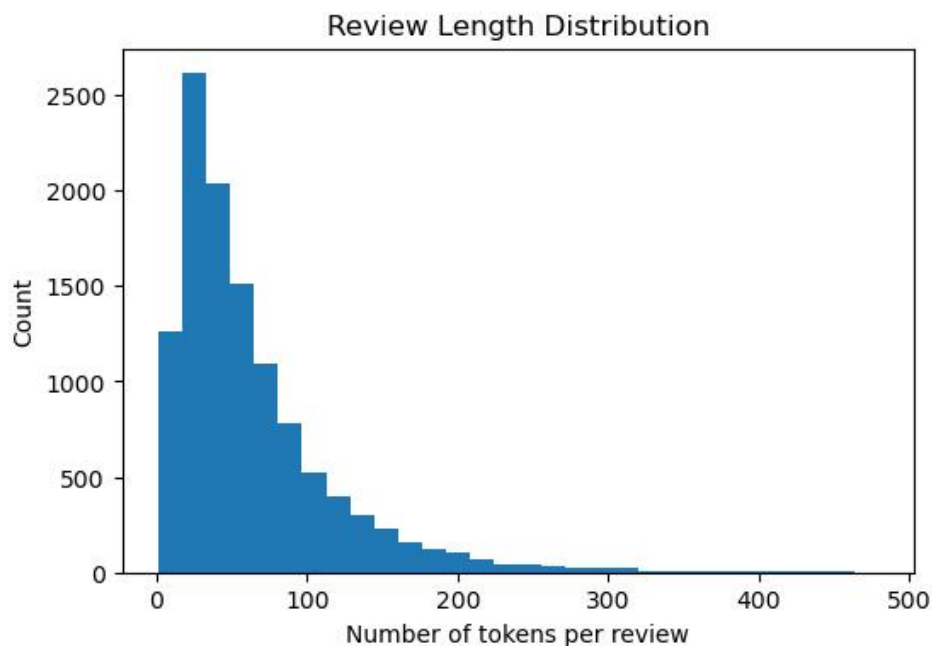
Over 1.2 million business attributes like hours, parking, availability, and ambience

Aggregated check-ins over time for each of the 138,876 businesses

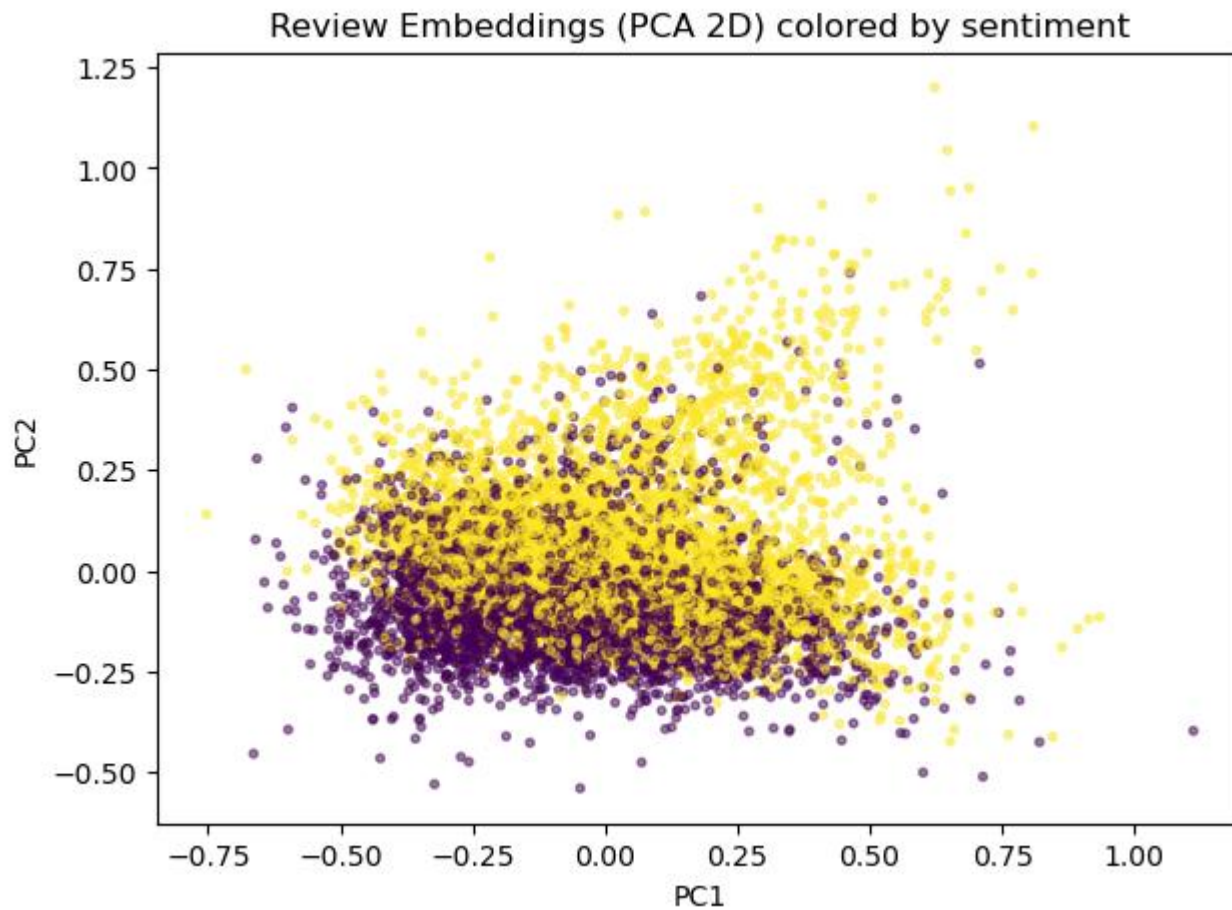
# Case Study

- <https://www.yelp.com/dataset>

|   | review                                            | sentiment |
|---|---------------------------------------------------|-----------|
| 0 | leave table feel ambivalent meh others would p... | 0         |
| 1 | time worth stick magical number three okay let... | 0         |
| 2 | love place boyfriend celebrate anniversary gla... | 1         |
| 3 | lovely even last night table great selection w... | 1         |
| 4 | table classic bistro good menu wine list cockt... | 0         |



# Case Study



# The End

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■ THANK YOU

Any questions?

